The illusion of hidden personal data

Project Report

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Executive Summary

Context and Objectives

To support a highly targeted marketing campaign for a new restaurant in the San Francisco Bay Area, this project identifies the five most influential accounts within a provided sample of a professional social network. Because the target audience is strictly limited to the Bay Area, our primary challenge was twofold : accurately inferring the geographic locations of users with undisclosed information, and mapping the spread of influence through the network to maximize regional reach.

Methodology

We approached this problem using a two-step data mining process guided by the CRISP-DM framework:

- **Location Prediction:** We developed a scoring algorithm that analyzes up to second-degree network connections to accurately predict whether a user is located in the San Francisco Bay Area or elsewhere.
- **Influence Modeling:** Treating information spread similarly to an infection propagation model, we used a Mean Field simulation to estimate the probability of users seeing and sharing promotional content. To find the optimal combination of five influencers out of billions of possibilities, we utilized a Greedy Algorithm optimized with a Tabu Search metaheuristic to avoid local maxima.

Key Findings

- **Location Inference:** Our prediction model successfully categorized the network, identifying **152** potential target users within the Bay Area with an estimated accuracy of **63%** against our ground-truth validation set.
- **Influencer Reach:** By running our optimized search simulation with a baseline sharing probability of 10%, we identified a combination of five specific influencers who collectively generate an exposure rate of **91%** across the Bay Area predicted target audience.

Recommendations

We recommend that the marketing department partner with the following five accounts: [**'U7024', 'U22747', 'U27287', 'U16141', 'U3955'**]. While there are inherent risks regarding prediction uncertainties and potential false positives, our simulation shows that this specific combination offers robust reach for the Bay Area campaign.

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1 Introduction

1.1 Problem Statement

The marketing department has tasked us with identifying the five most influential accounts to promote a restaurant located in the San Francisco Bay Area. This marketing campaign must be highly targeted, as the promotion is only relevant to individuals residing in this specific geographic region.

To achieve this, we have been provided with a sample of the LinkedIn social network. However, the dataset is currently incomplete, with only a fraction of the users having shared their geographic location.

1.2 Objectives

We have established the following primary objectives for this project:

1. **Location Prediction:** Determine a predicted geographic location for each user with missing data based on their network connections.
2. **Influence Modeling:** Model how information propagates through the network to determine the potential reach of any given user or combination of users.
3. **Influencer Selection:** Identify the optimal combination of five influencers that maximizes the campaign's reach within the target audience (the Bay Area).

1.3 Scope and Constraints

The scope of our project is specifically tailored to the provided network sample and focuses exclusively on maximizing reach within the San Francisco Bay Area.

Our analysis faces several constraints: primarily, the significant proportion of missing location data, which requires us to rely heavily on inferred geographical positions. Additionally, we are constrained by computational limits : the vast number of possible node combinations makes an exhaustive search for the optimal influencers impossible, requiring the use of heuristic approximation algorithms. Finally, our influence propagation assumes a simplified, uniform user behaviour that might not perfectly reflect reality.

1.4 Methodology Note

This project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework. This structured approach guides our analysis through distinct phases: understanding the business context and objectives, exploring the network data, modeling location predictions and influence propagation, evaluating the results against our business goals, and finally, recommending actionable strategies.

1.5 Report Structure

The remainder of this report is organized as follows: Section 2 details our Business Understanding and defines the specific success criteria. Section 3 covers Data Understanding, including exploratory analysis and data quality assessment. Section 4 details our Modeling approaches for both location prediction (Task 1) and influencer identification (Task 2). Section 5 presents the Evaluation of our models and discusses inherent limitations and risks. Finally, Section 6 provides our Conclusion and actionable recommendations for the marketing department.

2 Business Understanding

In this section we will analyse the subject, regarding the business context and the technical resources.

2.1 Business Goals and Context

As a data scientist at Linkedin, our aim is to find the 5 most influential people on the network who would be best suited to promote a restaurant in the Bay Area of San Francisco.

2.2 Translating Business Goals into Data Mining Goals

First, we need to define what an influencer is.
We decided that the best influencer in the Bay Area would be the person with the most direct and indirect contacts in the Bay Area. Each person would be given an influence score using a percentage of potential retweets to account for indirect contacts.

Therefore, we need to fill in the missing locations in the dataset, at least find the people who are in the Bay Area, as we are not interested in the other locations.

2.3 Success Criteria

The success criterion will be the percentage of people actually located in the Bay Area who have been reached (bearing in mind that some people will be reached at a rate of less than 100 % if they are not direct neighbors of an influencer).

3 Data Understanding

3.1 Data Description

We have a graph representing 811 users connected to each other by edges describing their LinkedIn connections.

In addition to this, there are three datasets: `location`, `college` and `employer` that describe the situation of each user in the graph. For this challenge, 60 % of the nodes were randomly and independently erased from the three datasets.
Each user may have had several employers and colleges but can only have one location.

Attribute	Type	Description	Data Availability
`name`	string	individual identifier	41.43 %
`location`	string	current user location	41.43 %

Table 1: `location` dataset structure

Attribute	Type	Description	Data Availability
`name`	string	individual identifier	28.36 %
`college`	string	user's current or former college	28.36 %

Table 2: `college` dataset structure

Attribute	Type	Description	Data Availability
`name`	string	individual identifier	36.62 %
`employer`	string	user's current or former employer	36.62 %

Table 3: `employer` dataset structure

The data is clean, with no duplicates or missing values.

3.2 Network analysis

Graph characteristics

Here is a table of the graph's characteristics :

Characteristic	Value
Number of nodes	811
Number of edges	1597
Density	0.0049

Table 4: Social graph description

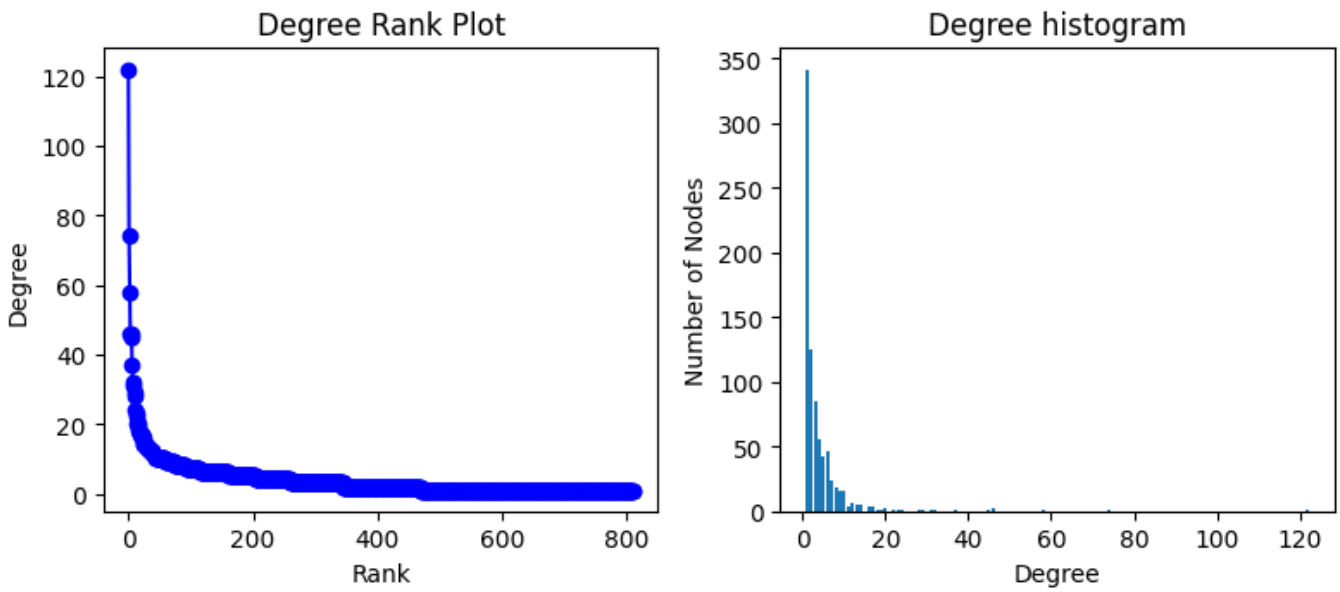

Figure 1: Graph density visualization

Geographic Distribution of Labeled Users

Here is an overview of the graph with the locations (from the complete location dataset) colored ("San Francisco Bay Area" is green): (see Figure 2)

We can see on the figure that there are few isolated San Francisco-located nodes that will be hard to predict.

Moreover, no node with a missing location has direct or second-degree neighbours with the same employer or college, so it will be difficult to use this information to fill in the missing locations unless we search for the missing college and employer values, which we will not do. In fact, completing them would be very uncertain given the low probability of two people working for the same employer or having attended the same college.

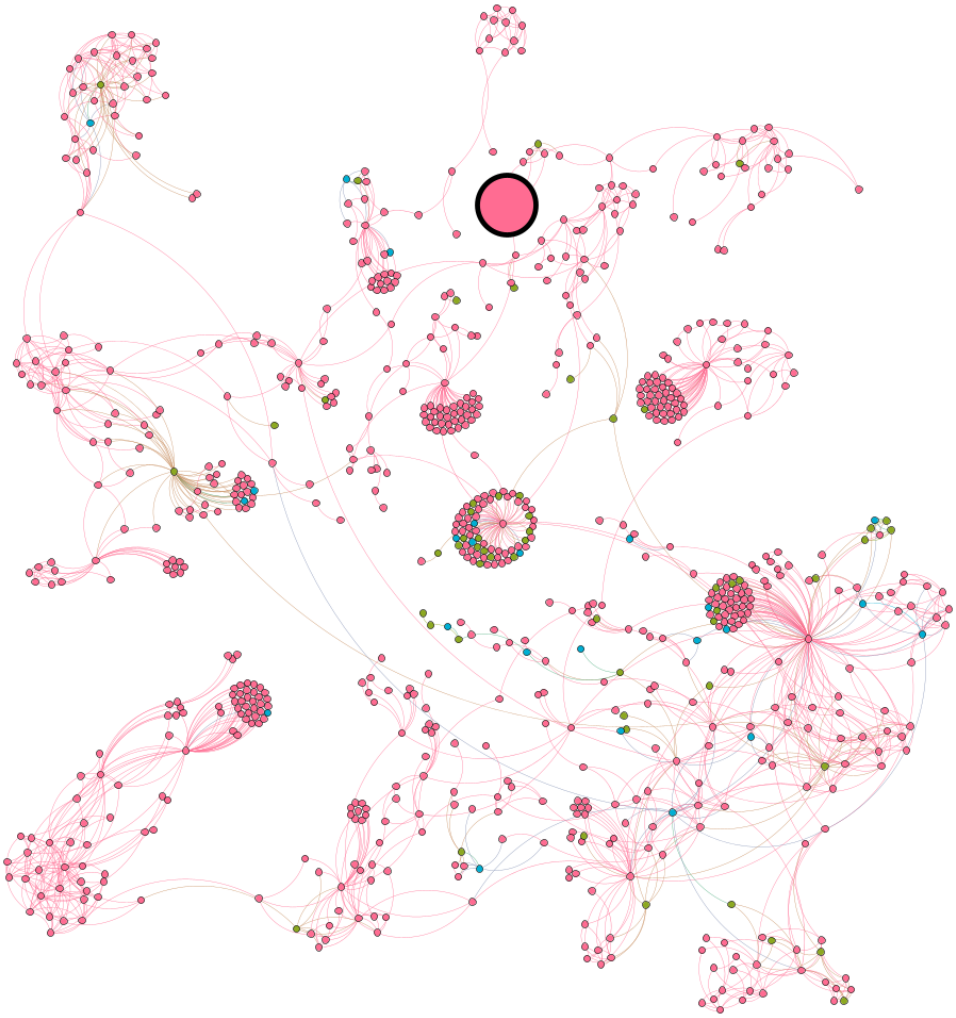

Figure 2: Visualization of all locations in the complete dataset on the graph (green = San Francisco missing value, blue = San Francisco not missing, and pink = other location)

4 Modeling

4.1 Task 1 — Location Prediction

Hypothesis

In order to fill in the missing location, we assume the following hypothesis:

1. People who work or have worked in the same company are more likely to live in the same area than other people in their network.
2. Filling in the location in the dataset progressively propagate a bias. And the result will depend on the starting node and the order of visit, so it's better to not consider the previous found location to find the next one.
3. A person is more likely to be in the same city as a direct connexion than an indirect one.
4. We want to maximize the recall metric because we want to touch a maximum of people in the Bay, even if this means considering too much users.

The model

The model is based on a scoring system designed to deduce users' locations by analyzing their local connections.
It operates by iterating through every node that currently lacks location data and scanning its social circle up to two degrees of separation to find any contacts who do have a registered location. When the algorithm identifies these connections, it calculates a **cumulative score** for each potential location based on the nature of the relationship.

A **direct neighbor** provides a base increase of one point to their specific location's score, while a **second-degree neighbor** (a friend of a friend) adds a weighted value labeled as `coef_1_2`, which is a tuned parameter between zero and one (we will describe how we choose it in a later section).

Furthermore, the algorithm applies a **specific geographical bias**: if a neighbor is located within the San Francisco Bay Area, an additional coefficient called `score_sf` is added to that location's tally to give it priority over other regions (we will describe how we choose it in a later section).

Once all nearby relations have been processed, the algorithm selects the location with the highest final score as the predicted residence for the node. However, it applies a final classification filter, where only the **`"San Francisco Bay Area"`** is kept as a specific label; if any other geographical area wins the scoring battle, the node is simply assigned the generic label **`"Other"`**. In cases where a node has absolutely no connections with a known location within two degrees, the algorithm defaults to a status of **`"NotFound"`**.

Choice of `coef_1_2` and `score_sf`

To find the most optimized `coef_1_2` and `score_sf` parameters, we varied these two parameters while observing the metrics obtained.
For variations in `score_sf`, there is no point in looking beyond 1 because after that the metrics stabilize, and the bias would be too significant compared to the weights given to each neighbor, which we set equal to 1.
As for `coef_1_2`, we vary it between 0 and 1 according to the assumptions.

Figure 3: Search for the most optimized parameters `coef_1_2` and `score_sf`

Based on the results shown in the figure, the most optimal pair of parameters, i.e., those that maximize recall while maintaining accuracy values above 60% and precision values above 30%, are `coef_1_2 = 0.2` and `score_sf = 0.4`, with:

```
Accuracy :  63.45%
Recall :  78.12%
Precision :  34.67%
```

Ineffective approaches

We also tested two other approaches to improve the model, which proved to be too complex or ineffective.

- First, we attempted to assign greater weight to relationships with individuals who had or currently had the same employer in order to validate the first hypothesis. However, this proved unnecessary since none of the neighbors, even second-degree neighbors, of individuals without a location had the same employer. This method could be effective on larger graphs.
- Secondly, we tried to complete the localizations found as the research progressed. However, the scores obtained were inconclusive because they were worse than those obtained previously. This allowed us to validate the second hypothesis, which stated that this practice propagated a bias.
- Finally, we attempted to analyse the cycles around the nodes we were searching for, assuming that if a cycle of up to five people existed with the same location, there was a greater chance that the location we were looking for was that one. However, the complexity of this search was such that our computers were unable to complete it. This avenue therefore remains to be explored.

4.2 Task 2 — Influencer Identification

Now that we have each user's predicted location, it is time to evaluate how much we can influence the target audience.

Assumptions and Defining Influence

For this task, we will assume the following :

- Each non-influencer user whose relations have shared a post on the event is considered to have seen it. They then have a probability of "retweeting", or sharing this post. This probability (named `SHARE_PROBABILITY`) is supposed uniform and constant and which value will be discussed below.
- All posts, whether they are an original post by an influencer, or another influencer, or a "retweet" of one, is considered as the same. This means an user can only "see", or "not see", and only "share" or "not share". (Since we are running probabilistic experiments, theses tags will not be stored as booleans, but as values between 0 and 1, see below.)

Then, for each user we define a `share_likelihood`, corresponding to the probability of them posting, or sharing a relevant post, as such :

- Each influencer makes a post on the social network, which gives them a `share_likelihood` of 1
- Each non-influencer has a `share_likelihood` equal to the probability of them seeing a post (or `exposure`, see below) from any of their relations, multiplied by their probability of sharing it.

We also define the probability for each user seeing a relevant post, or `exposure` as :

- Each influencer has an `exposure` of 1.
- each non-influencer has an `exposure` equal to 0 if the user is isolated, or :

$$\texttt{exposure}(U) = 1 - \prod_{V \in \texttt{neighbours(U)}} (1 - \texttt{share_likelihood}(V)$$

Therefore, our goal is to find the five influencers, which maximise (for a given `SHARE_PROBABILITY`) the function :

$$\texttt{influence_score}(\text{influencers}) = \frac{1}{|\texttt{target_users}|} \cdot \sum_{U \in \texttt{target_users}} \texttt{exposure}(U)$$

Simulations of the posts being shared by users

Our problem of exposure spreading through the different users is largely similar to an infection propagation model on a graph. This means we simply had to adapt and test some of the well known propagation algorithms. We finally used the **Mean Field algorithm**, for its rapidity, ease of implementation, and deterministic nature.

This algorithm works by iterations, updating each non-influencer node's `share_likelihood` each time with its neighbours' previous iteration values.

As seen in figure 4, the maximum delta rapidly reaches a plateau (around 0.003). This is likely due to the highly-clustered nature of the graph, making for small, cyclic oscillations around one value. However, this will not significantly impact results, as theses oscillations are relatively small.

Below is a visualisation of the influence of one random node (`SHARE_PROBABILITY` = 0.1).

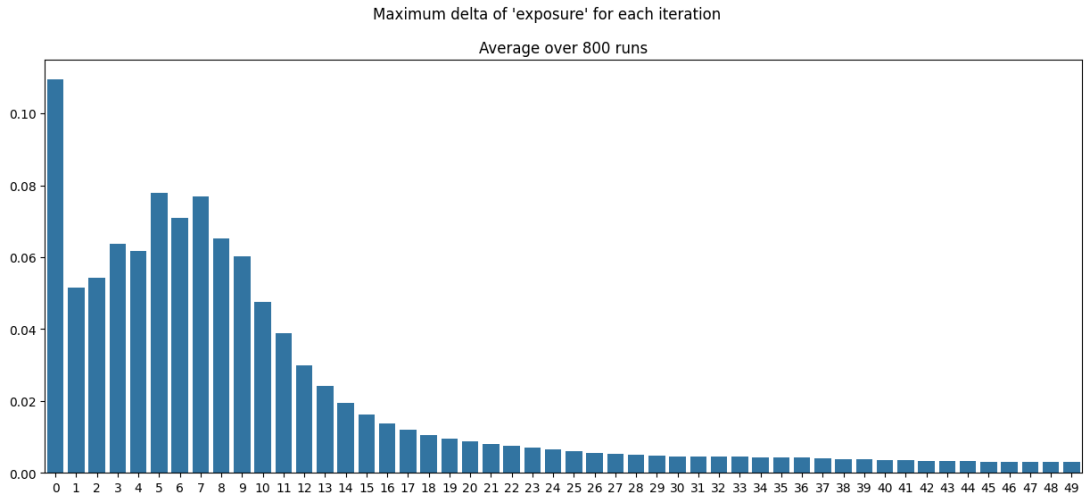

Figure 4: Evolution of the biggest change per iteration

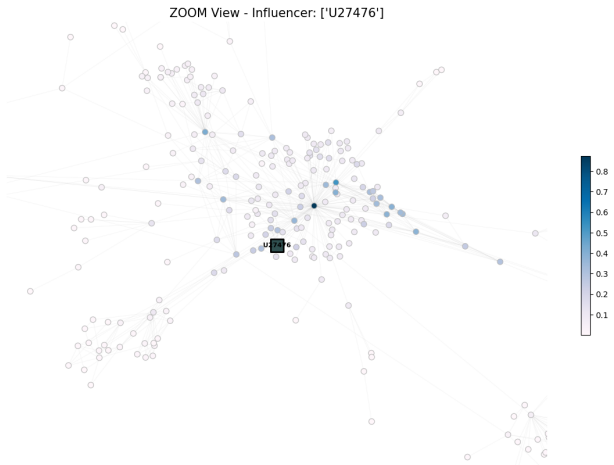

Figure 5: Exposure of one random influencer on his close relations

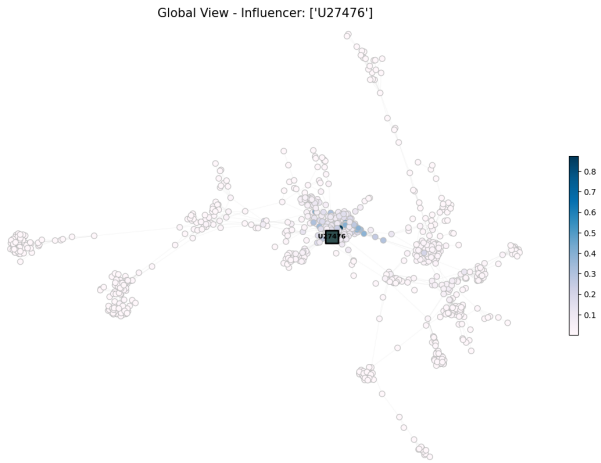

Figure 6: Global view of Figure 5

Searching for the best combination of influencers

The task of finding the absolute best influencer combination is also extremely hard. Exhaustive search, for our (relatively very small) dataset would mean running our simulation $\binom{811}{5} \simeq 2.9 \cdot 10^{12}$ times. A rather long run indeed. We therefore tried two approaches to tackle this problem.

a) Greedy Algorithm

Simple is often better. The greedy algorithm operates over five iterations, each adding one more user. At each iteration, every candidate user not yet included in the current selection is evaluated by running the simulation with that candidate added to the previously selected set. Once all candidates have been evaluated, the user yielding the best result is added to the selection. This process is repeated until five users have been selected.

This makes the algorithm run fast, only $811 + 810 + 809 + 808 + 806 = 4045$ runs. Much more manageable, this **takes around two minutes** on our computers. If this was too long (e.g. with a bigger social network sample), we could have implemented CLEF. It is an optimisation of the greedy algorithm that prioritizes previous bests, and exploits the fact that each node added influence can only go down. Still, this was not necessary here.

However, the risk of local optimum is quite large, so we also tried the following algorithm :

b) Tabu search

The tabu search algorithm is a classic metaheuristic specifically designed to escape the local optima we risk hitting with the greedy approach.

Instead of building from scratch, we use the 5 influencers found by our greedy algorithm as a starting point. At each iteration, we explore the "neighbourhood" of our current solution by swapping exactly one selected influencer with an unselected user. To evaluate these new combinations, we once again run our Mean Field simulation. As testing all 5×806 possible swaps at each step would take far too long, we simply evaluate a random subset of these neighbours.

The main advantage of this approach is the **tabu list**. When an influencer is swapped out, they are placed on this list for a fixed number of iterations. While on the list, it is strictly forbidden to add them back into our selection.

This mechanism forces the algorithm to explore entirely new branches, even if it means accepting a swap that temporarily lowers our objective function, `influence_score`.

By allowing these occasional backward steps, the algorithm can successfully climb out of local maxima. We let it run for a set number of iterations, keeping track of the absolute best combination found. While this obviously takes longer to run than the greedy approach, it gives us much more confidence that we haven't just settled for a local peak.

5 Evaluation

5.1 Influencer Selection Review

Alignment with Business Success Criteria

The primary objective of this project was to identify a group of five influencers capable of maximizing promotional reach within the San Francisco Bay Area. Our selection directly aligns with the established success criteria through the following results:

- **High Exposure Rate:** The optimized combination of five influencers (U7024, U22747, U27287, U16141, and U3955) generates a collective exposure rate (or `influence_score`) of 91% across the predicted target audience.
- **Probabilistic Reach:** By utilizing a baseline sharing probability of 10% (`SHARE_PROBABILITY` = 0.1), we identified accounts that effectively trigger information propagation beyond their immediate neighbors, ensuring deep penetration into the local network.
- **Target Precision:** The selection process prioritized nodes predicted to be in the Bay Area, ensuring that the marketing budget is focused on the relevant geographic region despite the initial 60% missing location data.

Tabu Search Impact

During testing, we finally discovered that the greedy algorithm did a "good enough" job, making useless the use of tabu search. This algorithm is extremely competent with getting out of a local optima, however this appears not to have been necessary.

SHARE_PROBABILITY Value Discussion

The value of `SHARE_PROBABILITY` is, as previously mentioned a difficult task, as we need to find a value that accurately simulate real world interactions on social media. However, using real world values is perhaps not ideal, as the relation graph we have is extremely small, with relatively very few nodes and edges.

Below is the average influence of one node on the whole graph :

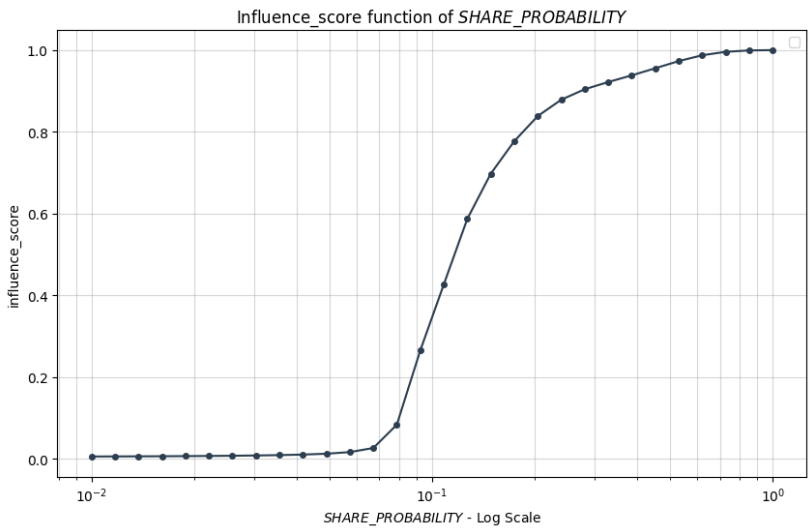

Figure 7: Average influence of one node, function of `SHARE_PROBABILITY`

The goal is to find value with which post reach parts of the network, but not all of it. This is why we have chosen a `SHARE_PROBABILITY` of **0.1**, which makes the most influential node have an `influence_score` of around 0.5 .

Moreover, after running the whole selection algorithm with different `SHARE_PROBABILITY` values, we obtain the figure 8 :

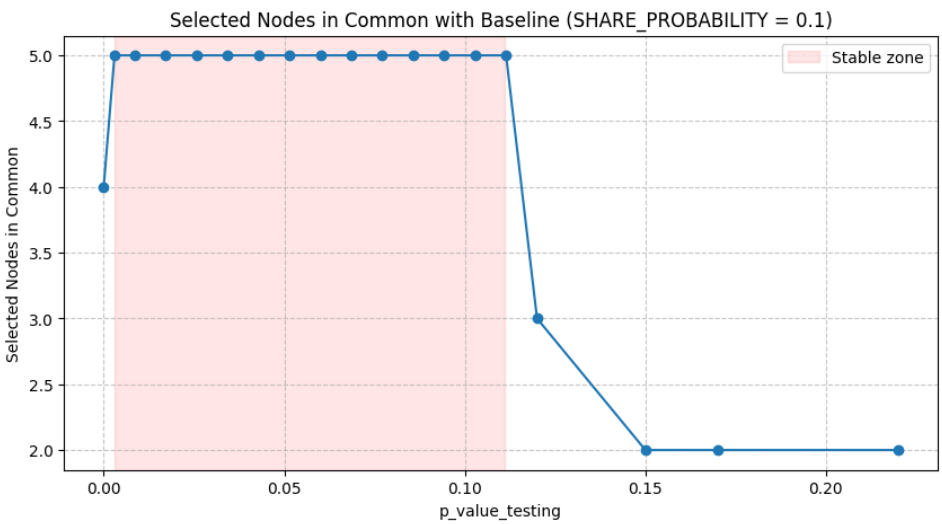

Figure 8: Selected nodes in common with baseline

We can observe that our algorithm predicts the same Influencer combination for `SHARE_PROBABILITY` in the range $[0.005; 0.11]$. This means that our algorithm is relatively resilient to this choice. However, for values greater than 0.13 we rapidly reach a new stable state. We did not choose this state as we believed that a lower `SHARE_PROBABILITY` was closer to reality. Still, the second stable state is `['U22747', 'U16128', 'U24095', 'U2691', 'U7024']`.

Sensitivity to Prediction Errors

Since we are not sure of all predictions made in **Task 1** (see 4.1), we also evaluated the `influence_score` of the selected influencers, but with only the nodes that explicitly have declared living in the target location.

5.2 Limitations and Risks

Prediction uncertainty Propagation

Since our influencer selection is performed on a "augmented" dataset containing both ground-truth and predicted locations, the final influence scores are sensitive to the performance of the prediction model. We identified two primary risks:

- **The False Positive Risk (Ghost Reach):** If the model predicts a user is in the Bay Area when they are actually in "Other," the influencer's calculated reach will be artificially inflated. In a real-world scenario, this would result in marketing spend being wasted on users who cannot visit the restaurant.
- **The False Negative Risk (Hidden Opportunity):** Conversely, if the model labels a Bay Area resident as "Other" or "NotFound," we may undervalue an influencer who actually has a high density of followers in the target zone.

Potential biases Induced by our Assumptions

a) Homogeneity discussions

We have assumed that users who have decided to hide their personal information are homogeneous in the population, however, in real case scenarios, they might be part of social groups (age, privacy concerns...) which in turn might lead to bias in the prediction.

b) More is not always better

We have assumed that all relations are of equal nature, and lead to a single probability of "retweet". This is easily refutable, some users might have a very high number of relations, but of poor quality, whereas some other might have a small, but very engaged community.

6 Conclusion and Future Work

6.1 Summary of Findings

This study demonstrates that a highly targeted marketing campaign can be executed even when facing significant data gaps in a professional social network.

- **Location Inference:** We successfully developed a scoring algorithm that analyses network connections up to the second degree. This model identified 152 potential target users in the Bay Area.
- **Optimal Parameters:** Testing revealed that the most effective balance for location prediction was achieved with `coef_1_2` $= 0.2$ and `score_sf` $= 0.4$, resulting in a recall of 78.12% and an accuracy of 63.45%.
- **Propagation Modeling:** Using a Mean Field simulation, we observed that the network's clustered nature allows for rapid information spread, though it is subject to small cyclic oscillations.
- **Algorithm Efficiency:** While we explored complex metaheuristics like Tabu Search, evaluation showed that a Greedy Algorithm provided a "good enough" solution for this network size, finding a solution achieving 91% with our target metric.
- **Final Recommendation:** The identified influencer set (['U7024', 'U22747', 'U27287', 'U16141', 'U3955']) offers a robust and actionable strategy for the San Francisco Bay Area campaign.

6.2 Top 5 Influencer Recommendations

The final influencer selection is :

```
['U7024', 'U22747', 'U27287', 'U16141', 'U3955']
```

The coverage graph is below :

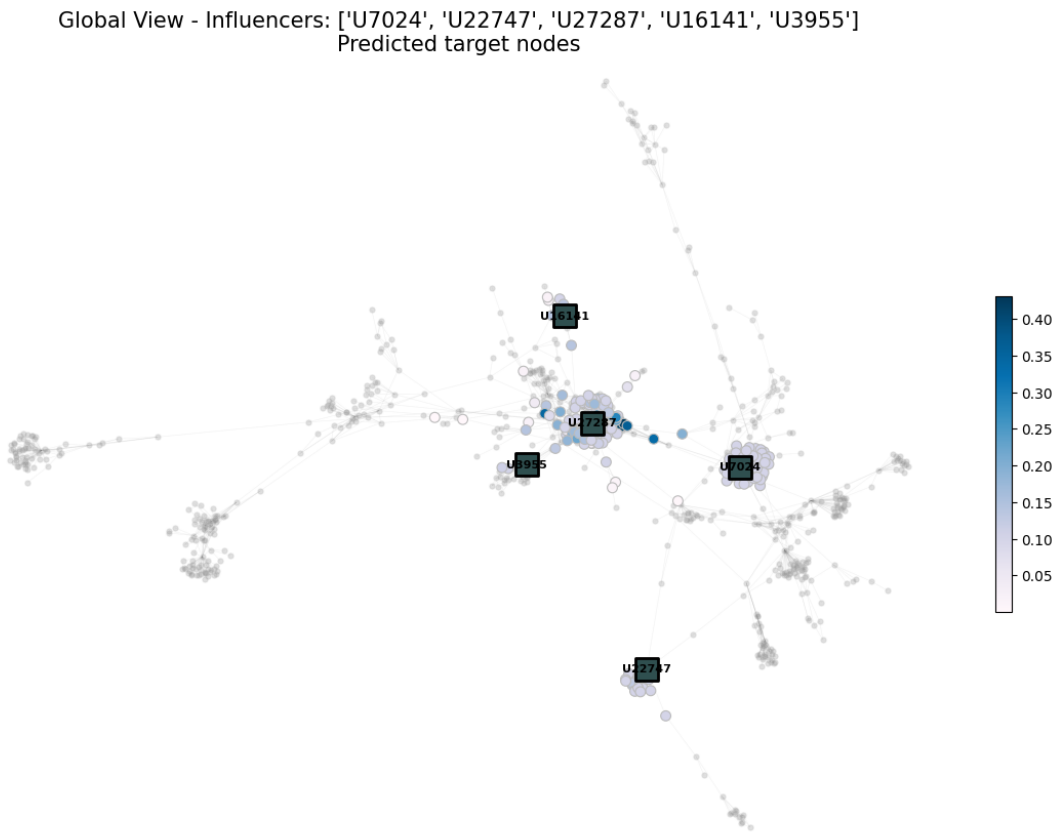

Figure 9: Influence of the final influencer selection on the predicted target audience

6.3 Scalability and Deployment Considerations

While our algorithms run in a manageable time with the given network, a much bigger one would become difficult to manage. We tried to keep algorithmic complexity under control, ($O(N)$, N being the number of users), but real social networks are orders of magnitude bigger.

6.4 Future Improvements

More scalable algorithms

To continue on section 6.3, use in an actual production environment would require the use of much more optimised algorithms, such as *Reverse Influence Sampling*. This works by going backwards from an homogeneous subset of the target audience, and finding the most influent sources.

Limiting the impact of prediction errors

The impact of prediction errors, talked about in part 5.2 can be limited through :

- Since we know which nodes are predicted and which are not, we can, for the value given to the influencer selection algorithm, add weights to make sure to give more importance to users we are certain are in the target area. This could also be extended to include the confidence of the prediction.